

SEMESTER S6

SEMICONDUCTOR DEVICE MODELING

Course Code	PEEVT 636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Solid State Devices (PCECT302)	Course Type	Elective

Course Objectives:

1. This course explains the qualitative understanding of the physics of the semiconductor devices and conversion of this understanding into equations
2. To introduce students to the field of compact modeling and familiarize them with the methods for the model development

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Semiconductor device modeling overview: Types of device models- Numerical Models, Analytical models, compact models. PN Junction diode: DC Current-Voltage Characteristics, Limitations of the ideal diode model, Static model of the ideal diode and its implementation in SPICE2, Static model of the real diode and its implementation in SPICE2, Large-signal model of the diode, Small-signal model of the diode, Temperature and area effects on the diode model parameters.	9
2	Bipolar junction transistor : Ebers-Moll static model, Ebers-Moll large signal mode , Ebers-Moll small signal model, Gummel – Poon static model, Temperature and area effects on the BJT model parameters. Heterojunction Bipolar junction transistor (HBT): SiGe HBT, Static operation and characteristics, HICUM level 2: Internal transistor model.	9
3	The four terminal MOS structure: strong inversion, complete symmetric strong inversion model, simplified symmetric strong inversion model, simplified source referenced strong inversion model. MOS Transistor SPICE models: Level 1 static model, level 2 static model,	9

	level 1 and level 2 large signal model, small signal model, effect of temperature on the model parameters. MOSFET BSIM models: BSIM1 model, parameter listing, BSIM1 static model, BSIM1 charge model.	
4	MOSFET BSIM models: BSIM1 model, parameter listing, BSIM1 static model, BSIM1 charge model. Advanced semiconductor device modeling: Tunnel FET: need of TFET, device structure, TFET modelling approach, modelling the surface potential source channel junction: pseudo 2D method, modelling drain current: constant polynomial term assumption, tangent line approximation. Modeling of novel unipolar devices: Device structures and principles of operation of FinFET, HEMT, TFET. Basic modeling concepts-	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement the static model of an ideal diode and a real diode in SPICE2 software, demonstrating the ability to set up and run simulations.	K3
CO2	Implement and simulate the Ebers-Moll and Gummel-Poon static models for BJTs in SPICE2, demonstrating practical application skills.	K3
CO3	Explain the differences between the complete symmetric strong inversion model, the simplified symmetric strong inversion model, and the simplified source-referenced strong inversion model for MOS structures.	K2
CO4	Explain the TFET modeling approach and the importance of modeling the surface potential at the source-channel junction.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		1							
CO2	3	2	2		1							
CO3	3	2	2		1							
CO4	3	2	2									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	COMPACT HIERARCHICAL BIPOLAR TRANSISTOR MODELING WITH HICUM	Michael Schroter	World Scientific	1/e, 2010
2	Semiconductor Device Modeling With SPICE	Giuseppe Massobrio	McGraw-Hill	2/e, 1993
3	Tunnel Field-Effect Transistors (TFET) Modelling and Simulation	Jagadesh Kumar Mamidala	John Wiley	1/e, 2017
4	Operation and Modelling of the MOS Transistor	Yannis Tsividis	Oxford University Press	3/e, 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	SEMICONDUCTOR DEVICES MODELLING AND TECHNOLOGY	Nandita Das Gupta	PHI learning	1e/2004
2	Semiconductor Device Modelling	Christopher M. Snowden	Springer	1e/2012
3	Semiconductors Physics and Devices	Donald Neamen	Tata Mc Graw Hill	4e/2017
4	Physics of Semiconductor Devices	S M Sze	Wiley	3e/2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117106033
2	https://nptel.ac.in/courses/117106033
3	https://nptel.ac.in/courses/117106033
4	https://www.youtube.com/playlist?list=PLbMVogVj5nJQ2k2HAGHFENB4a2nI8OppN